

Generalizing Transfer Bayesian Optimization to Source-Target Heterogeneity

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Abstract—Black-box optimization algorithms typically start a search from scratch, assuming little prior knowledge about the task at hand. In practice, this approach can be prohibitive for computationally expensive problems, as a large number of costly function evaluations are often needed before a suitable (near-optimal) solution is found. Under this observation, recent efforts have incorporated *transfer learning* capabilities into sequential model-based Bayesian optimization (BO) solvers, resulting in substantial performance speed-ups by leveraging information from related past problems. However, a common simplifying assumption in existing approaches is that the search spaces of a previously encountered *source* and the ongoing *target* task bear the same features and dimensionality, with the difference lying in their respective objective functions. In this article, we present a generalized *transfer BO* algorithm that relaxes the aforementioned assumption. Our method jointly transforms source features while training probabilistic transfer regression models for the target, thus applying to practical use-cases where (in addition to the difference in objective functions) the number of features could change across the source and target tasks; for example, features can be added and/or removed. The theoretical basis of our proposal is analyzed, and its empirical performance is demonstrated on synthetic benchmark functions as well as in realistic examples spanning engineering design and the automated configuration of a machine learning model.

Note to Practitioners—Problems of industrial interest have a tendency of being repetitive in nature. For this reason, domain experts are always in high demand, as they are able to harness their experience of similar problems to come up with fast solutions in difficult situations. However, domain experts are not easy to find. Given this fact, the present paper puts forth

a method for automating the process of knowledge extraction (through experiential learning) and transfer across problems in the domain of computationally expensive black-box optimization. The key novelty and motivation of this work lies in enabling the adaptive transfer of knowledge even when the number of features changes across the source and target problems. Our proposed approach is verified experimentally on a range of benchmarks as well as real-world problems of a computationally expensive nature, highlighting the utility of an optimization engine that is able to learn from experience without the need for constant human intervention.

Index Terms—Bayesian optimization (BO), expensive global optimization, Gaussian process (GP), transfer learning.

I. INTRODUCTION

REAL-WORLD problems seldom exist in isolation. Any practically useful system in an industrial setting is expected to tackle a large number of similar problems over a lifetime, all of which display domain-specific idiosyncrasies. Taking this cue, in the domain of machine learning, the idea of exploiting training signals from various *source* tasks to improve the learning on a related *target* task has seen considerable success under the label of *transfer learning* [1], [2]. In contrast, it is observed that similar efforts in the domain of *optimization* have been scarce, with most solvers starting the search from scratch assuming a zero prior knowledge state. It can be argued that for many real-world problems, especially those entailing high cost of evaluations, overlooking useful information contained in data drawn from related tasks [3] may lead to deleterious computational overheads in the reexploration of overlapping search spaces.

For dealing with computationally expensive black-box optimization problems, Bayesian optimization (BO) has emerged at the forefront due to its high sample efficiency [4]. For a given task, the method serves as a sequential model-based optimization approach that iterates over the following steps until the available computational budget is exhausted: 1) a data set of evaluated solution samples is used to learn a probabilistic regression model $p(f)$ of the objective function f [typically in the form of a Gaussian process (GP)]; 2) $p(f)$ is used to induce a relatively cheap acquisition function that leverages the uncertainty in the posterior to guide exploration; 3) the cheap acquisition function is optimized to determine the next solution to evaluate; and 4) the regression data set is augmented with the newly evaluated solution sample.

Regardless of its well-known sample efficiency, a challenge BO continues to face is that of *cold start*, indicating the difficulty of gathering enough data to build a sufficiently

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informative probabilistic model that can effectively guide the search. To address this issue, recently proposed methods have shed light on the potential advantages of incorporating transfer learning capabilities into the optimizer [5], [6]; for instance, reusing data generated from related source problems to improve the accuracy of the target regression model. While instilling such experiential learning capabilities into BO has shown promising results, we note that most existing methods are still limited in the extent of their applicability. A common assumption in the literature is that the search spaces of the source and target tasks have the same features and dimensionality, with the difference between them lying in their respective objective functions. While some applications of knowledge transfer in BO are consistent with this assumption, there exist many others where the number of features could change across the source and target tasks.

Given the above, in this article, we propose a novel transfer Bayesian optimization (TBO) algorithm that relaxes the limiting assumption of existing methods. Our framework generalizes TBO to settings characterized by source-target heterogeneity, encompassing practical use-cases where (in addition to the difference in objective functions) features could be added and/or removed in an ongoing target problem relative to those present in the source. To automate the utilization of latent correlations arising from the subset of overlapping features, we consider *jointly* transforming the source data while training probabilistic transfer regression models for the target. To this end, a nonlinear feature transformation layer is introduced into the transfer covariance function of an adaptive transfer GP (TGP) model [7], [8], mapping the source to a new representation that is aligned with that of the target task. The applied transformation can either be randomized (in the simplest case), hand-crafted by the optimization practitioner (using domain knowledge), or learned from available data (in the most general case). Our theoretical regret bound analysis indicates that incorporating the resultant heterogeneous TGP (Hetero-TGP) model into TBO, to form what we refer to as the Hetero-TBO algorithm, guarantees sample efficiency superior to that of conventional BO without transfer learning.

Real-world applications of the Hetero-TBO algorithm are expected to be aplenty. In engineering systems, designs naturally evolve over time, with new design features being routinely added and redundant ones removed to help meet changing performance requirements [9]. While there exist definite relationships between the original and modified designs, the features of their respective search/parameter spaces are bound to change, providing an ideal setting for the unique facets of Hetero-TBO to come to the fore. While heuristic strategies for transferring knowledge across such nonhomogeneous design optimization problems have been studied [10], [11], the present work is, to the best of our knowledge, the first to take a principled BO approach in this regard. Our emphasis lies in solving problems of a computationally expensive nature, also spanning the automated configuration of machine learning (AutoML) algorithms [12], [13], a domain that has lately been attracting significant research attention.

A detailed exposition of the ideas discussed heretofore is organized in this article as follows. Section II provides a

brief description of classical BO, and an overview of recent works that have introduced the notion of knowledge transfer within BO. Next, in Section III, the generalized Hetero-TGP model is developed, which is then to be incorporated into the framework of TBO. The Hetero-TBO algorithm is described and theoretically analyzed in Section IV. In Sections V–VII, we numerically investigate the algorithm on a diverse test bed. Section VIII concludes this article.

II. BACKGROUND AND RELATED WORK

Without loss of generality, we consider the target optimization problem to be stated as the minimization task

$$\arg \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \quad (1)$$

where $\mathbf{x}$ is a d -dimensional solution vector in the search space $\mathcal{X} \subset \mathbb{R}^d$, and f is a computationally expensive black-box objective function. This implies that there is no closed-form expression for f , and one must query an oracle (e.g., a computer simulation) to obtain the value of f at a given $\mathbf{x}$. We denote the optimal value of $\mathbf{x}$ as $\mathbf{x}^{\text{opt}}$.

In conventional BO [4], the search begins by sampling and evaluating a small set of candidate solutions in $\mathcal{X}$. In subsequent iterations, the solutions to query are selected by maintaining a balance between *exploration* and *exploitation* of the search space, utilizing all information at hand with the goal of maximizing sample efficiency. Let the data set after N exact solution evaluations be $\mathcal{D}^{(N)} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$, where $y^{(i)} = f(\mathbf{x}^{(i)}) + \epsilon$ is the observed value of the possibly noisy objective function at input $\mathbf{x}^{(i)}$. It is noted that ϵ is assumed to be an additive i.i.d noise term defined as $\epsilon \sim \mathcal{N}(0, \sigma_n^2)$. Given $\mathcal{D}^{(N)}$, a probabilistic regression model $p(f|\mathcal{D}^{(N)})$, typically in the form of a GP, is learned. The predictive distribution of the GP is then used to define a cheap acquisition function $\alpha(\mathbf{x}; p(f|\mathcal{D}^{(N)}))$ quantifying the figure of merit for a solution $\mathbf{x}$ to be considered for evaluation. With this, active selection of the next solution to evaluate is carried out by solving the following:

$$\mathbf{x}^{(N+1)} = \arg \max_{\mathbf{x} \in \mathcal{X}} \alpha(\mathbf{x}; p(f|\mathcal{D}^{(N)})). \quad (2)$$

The augmented data set $\mathcal{D}^{(N+1)} = \{\mathcal{D}^{(N)}, (\mathbf{x}^{(N+1)}, y^{(N+1)})\}$ is used for updating the regression model, guiding the subsequent round of solution selection. The process continues until some predefined stopping condition, such as the exhaustion of available computational budget, is satisfied.

A. State-of-the-Art Advances in Transfer Optimization

The concept of knowledge transfers in optimization, alternatively referred to as *transfer optimization*, has been recently introduced as a way to reproduce high performance by learning across problems and domains [14], [15]. Roughly speaking, for any class of algorithms, if an efficient mechanism for identifying and reusing relevant source information can be devised, then the performance of the resultant transfer optimizer should be no worse than a standard solver tackling problems from scratch. In [16]–[18], this claim is substantiated both theoretically and empirically for a class of search distribution

modeling-based transfer evolutionary algorithms. The boost in evolutionary search performance is facilitated by an online learning module minimizing the threat of *negative transfers* (i.e., harmful intertask interactions), spanning even black-box optimization settings where source-target similarities are not known beforehand.

While the aforementioned papers mark significant strides in realizing the transfer optimization paradigm, they are focused on population-based search methods that are not ideally suited for computationally expensive problems. Accordingly, in this article, we turn our attention to TBO algorithms. In BO, Yogatama and Mann [19], Bardenet *et al.* [20] proposed to incorporate prior knowledge through the transfer of GP covariance functions across related tasks. The underlying assumption is that different problems often possess *similar looking* objective functions (albeit with different location and scale parameters), thus encouraging the reuse of data under appropriate output normalizations. The normalization is achieved via *z-scores* of the observed target data in [19], while scale-independent *ranking* information is used for performing GP regression in [20]. In contrast to the direct transfer of GP covariance functions, a metalearning scheme explicitly biasing target evaluations toward optimized solutions of the most similar source problems was put forward in [5]. To ascertain the extent of source-target similarity, a regression model was trained offline to map numerical descriptors (i.e., manually defined metafeatures) of pairs of optimization tasks to the Spearman correlation between their respective objective functions. Higher the predicted correlation, greater is the expected source-target similarity.

A multitask GP was incorporated in [6] for building a joint model $p(f_{\mathcal{T}}, f_{\mathcal{S}} | \mathcal{D}_{\mathcal{T}}, \mathcal{D}_{\mathcal{S}})$ of the objectives of the source ($\mathcal{S}$) and target ($\mathcal{T}$) in multitask BO. A combined covariance matrix was constructed using a product covariance function, accounting for the relationship between inputs in conjunction with the relationship between tasks. A novel acquisition function was proposed for the multitask setting in [21], with further scalability enhancements achieved in [22] using Bayesian neural networks instead of GPs.

In [23], adaptive knowledge transfers from multiple sources to a single target optimization task was carried out via a multiproblem surrogate approach. A linearly weighted combination of all available source and target regression models was derived by minimizing the squared error of out-of-sample predictions on the target data. A similar methodology was also proposed in [24]; however, instead of minimizing the squared error, a ranking loss function was used to determine the weight of a model according to the probability of it being the best predictor in the ensemble. In [25], two alternatives for determining the aggregation weights were described. The first is the product of experts [26], and the second is kernel regression where weights are set as per an approximate similarity kernel between the source and target data sets.

Apart from the above, there exist a plethora of other transfer learning algorithms, including the *stacking* of TGPs [8], *boosting* for regression transfer [27], multiple kernel transfer learning [28], transfer Bayesian committee machines [29], and so on, that could offer unique advantages when applied in a

TBO framework. However, a common facet of the majority of these methods is that they assume the input features of the source and target tasks to be the same. Even in the case of GP-based transfer learning, the baseline TGP kernel becomes ill-defined if the dimensionalities of the source and target input spaces do not match. This hinders their applicability in cases of source-target heterogeneity, which are of primary interest in the present paper. We note that although heterogeneous transfer learning has been widely studied in the past for classification models, much less attention has been paid to regression [27]. Thus, to generalize TBO to source-target heterogeneity, in Section III we propose a novel Hetero-TGP regression model that involves joint source feature transformation as a form of domain adaptation. In the subsequent Sections IV–VII, we show that the Hetero-TGP model provides useful theoretical advantages as well as promising empirical performance.

III. GENERALIZED TGP

A standard GP is a stochastic nonparametric approach for regression that extends the concept of multivariate Gaussian distributions to infinite dimensions. Thus, a GP is completely specified by its mean and covariance functions, with a real process f typically written as [30]

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (3)$$

where $m(\mathbf{x})$ represents the distribution mean that is often set to zero, and $k(\mathbf{x}, \mathbf{x}')$ is the kernel (or covariance function) that relates any two observations $f(\mathbf{x})$, $f(\mathbf{x}')$. Given a data set $\mathcal{D} = \{X, \mathbf{y}\} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$, the kernel parameters are tuned to determine an appropriate GP prior, based on which the predictive distribution at a new input $\mathbf{x}^{(*)}$ is given by

$$p(f^{(*)} | X, \mathbf{y}, \mathbf{x}^{(*)}) \sim \mathcal{N}(\hat{\mu}(\mathbf{x}^{(*)}), \hat{\sigma}^2(\mathbf{x}^{(*)})) \quad (4)$$

$$\hat{\mu}(\mathbf{x}^{(*)}) = \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{y} \quad (5)$$

$$\hat{\sigma}^2(\mathbf{x}^{(*)}) = k_{**} - \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{k}_* \quad (6)$$

where $K(X, X)$ is the $N \times N$ Gram matrix whose ij th entry is $k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$, $\mathbf{k}_* = K(\mathbf{x}^{(*)}, X)$, $k_{**} = k(\mathbf{x}^{(*)}, \mathbf{x}^{(*)})$, and σ_n is additive noise; T indicates a transpose operator.

In the case of transfer learning, we consider a source domain $\mathcal{S}$, typically with a sufficient amount of data, to be used to improve the learning in a related target domain $\mathcal{T}$ in which there is only limited data. Let $\mathcal{D}_{\mathcal{S}} = \{X_{\mathcal{S}}, \mathbf{y}_{\mathcal{S}}\} = \{(\mathbf{x}_{\mathcal{S}}^{(i)}, y_{\mathcal{S}}^{(i)})\}_{i=1}^{N_{\mathcal{S}}}$ and $\mathcal{D}_{\mathcal{T}} = \{X_{\mathcal{T}}, \mathbf{y}_{\mathcal{T}}\} = \{(\mathbf{x}_{\mathcal{T}}^{(i)}, y_{\mathcal{T}}^{(i)})\}_{i=1}^{N_{\mathcal{T}}}$ denote the source and target data sets, respectively, with $N_{\mathcal{S}} \gg N_{\mathcal{T}}$. As proposed in [7], a TGP model can achieve adaptive knowledge transfer from $\mathcal{S}$ to $\mathcal{T}$, when $f_{\mathcal{S}} \neq f_{\mathcal{T}}$, by redefining the covariance function $k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})$ in the following manner:

$$\ddot{k}(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \begin{cases} \lambda k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}), & \mathbf{x}^{(i)} \in X_{\mathcal{T}} \text{ \& } \mathbf{x}^{(j)} \in X_{\mathcal{S}} \\ & \text{or } \mathbf{x}^{(i)} \in X_{\mathcal{S}} \text{ \& } \mathbf{x}^{(j)} \in X_{\mathcal{T}} \\ k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}), & \text{otherwise.} \end{cases} \quad (7)$$

Here, λ is a parameter capturing the source-target correlation, and $k(\cdot, \cdot)$ is any valid covariance function quantifying the relationship between inputs. If $-1 \leq \lambda \leq 1$, the matrix $\ddot{K}$

constructed by $\tilde{k}(\cdot, \cdot)$ is guaranteed to be positive semidefinite (PSD), implying that the TGP model is always valid under this sufficiency condition. However, for any covariance function to be usable in (7), the input spaces $\mathcal{X}_S$ and $\mathcal{X}_T$, from which X_S and X_T are drawn, should bear the same features and dimensions; i.e., $\mathcal{X}_S = \mathcal{X}_T$. Otherwise, $k(\cdot, \cdot)$ is ill-defined. This highlights a core limitation of the existing TGP model when it comes to applications characterized by source-target heterogeneity, thus motivating the source feature transformation mechanism as described next.

A. Hetero-TGP With Source Feature Transformation

In this article, we consider situations where, in addition to $f_S \neq f_T$, the number of features could change across the source and target tasks. That is, if $\mathcal{X}_S \subset \mathbb{R}^{d_S}$ and $\mathcal{X}_T \subset \mathbb{R}^{d_T}$, we have $d_S \neq d_T$, encompassing practical use-cases where features are added and/or removed in the target relative to those present in the source. At the same time, we consider the tasks to share a subset of common features, which gives rise to the potential for latent correlations between them.

Given the problem setting above, we introduce a feature mapping M for the purpose of aligning the source data with the target. Building on [31], we propose an approximation to the full Bayesian framework by defining M as a deterministic, parameterized function. In particular, the new source representation is induced via an l -layer neural network, where the i th layer performs the following transformation:

$$\psi_i(Z_i) = \phi(W_i \cdot Z_i^T + \mathbf{b}_i)^T. \quad (8)$$

Here, Z_i is the input of the i th layer, implying $Z_1 = X_S$ and $Z_{i+1} = \psi_i(Z_i)$; W_i is a matrix of synaptic weights, $\mathbf{b}_i$ is a bias vector, and ϕ represents a neural activation function. The overall transformation of the source data can then be written as the composition function

$$X'_S = M(X_S) = (\psi_l \circ \dots \circ \psi_1)(X_S). \quad (9)$$

M is *admissible* as long as matrix W_l has d_T rows and $\mathbf{b}_l$ is d_T -dimensional. The final layer of the neural network thus produces d_T outputs, which corresponds with the dimensionality of the input space of the target task.

Under the transformation of (9), the TGP covariance function of (7) is modified as

$$\tilde{k}(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \begin{cases} \lambda k(M(\mathbf{x}^{(i)}), \mathbf{x}^{(j)}), & \text{case A} \\ \lambda k(\mathbf{x}^{(i)}, M(\mathbf{x}^{(j)})), & \text{case B} \\ k(M(\mathbf{x}^{(i)}), M(\mathbf{x}^{(j)})), & \text{case C} \\ k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}), & \text{otherwise} \end{cases} \quad (10)$$

where case A := $(\mathbf{x}^{(i)} \in X_S; \mathbf{x}^{(j)} \in X_T)$; case B := $(\mathbf{x}^{(i)} \in X_T; \mathbf{x}^{(j)} \in X_S)$; case C := $(\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \in X_S)$. With this, the Gram matrix of the resultant Hetero-TGP model takes the following form:

$$\tilde{K} = \begin{pmatrix} K_{SS} & K_{ST} \\ K_{TS} & K_{TT} \end{pmatrix} \quad (11)$$

where K_{SS} and K_{TT} are the source-to-source and target-to-target block matrices, respectively, and $K_{ST} (= K_{TS}^T)$ is the

source-to-target block matrix. The sufficiency condition on λ for guaranteeing positive semidefiniteness of $\tilde{K}$ holds under the Hetero-TGP kernel as well.

There are various ways in which the parameters of the neural network mapping function M can be determined. In the simplest case, they may be randomly assigned, relying on parameter λ to adapt to the intertask similarities emerging from the random transformation. Unsupervised domain adaptation methods may also be considered. However, despite the cost of unsupervised learning, the learned data representation may not be useful for the target regression task [31]. Thus, a principled approach is to *jointly* learn the parameters of M together with the parameters of the TGP kernel in a fully supervised manner. In some cases, sufficient domain knowledge could be available for M to be hand-crafted. A trivial example occurs when the source and target tasks are known to have the same input features and data distribution, such that M can immediately be set to the identity function. As such, the baseline TGP can be thought of as a special case of the Hetero-TGP model.

B. Supervised Learning of the Hetero-TGP Model

Supervised training of the full Hetero-TGP model involves joint optimization of the parameter set

$$\theta = \{\theta_{\tilde{k}}, \sigma_S^2, \sigma_T^2, \lambda, W_1, \dots, W_l, \mathbf{b}_1, \dots, \mathbf{b}_l\}. \quad (12)$$

Here, $\theta_{\tilde{k}}$ represents the parameters of the modified covariance function, and σ_S^2 and σ_T^2 are the source and target noise variances, respectively. In the TGP [7], the marginal likelihood of the target data is to be maximized conditioned on the available source data. We accordingly optimize for the following conditional distribution in the Hetero-TGP:

$$p(\mathbf{y}_T | \mathcal{D}_S, M, X_T) \sim \mathcal{N}(\boldsymbol{\mu}_T, C_T) \quad (13)$$

where

$$\begin{aligned} \boldsymbol{\mu}_T &= K_{TS}(K_{SS} + \sigma_S^2 I)^{-1} \mathbf{y}_S \\ C_T &= (K_{TT} + \sigma_T^2 I) - K_{TS}(K_{SS} + \sigma_S^2 I)^{-1} K_{ST}. \end{aligned} \quad (14)$$

Based on (13), the log-likelihood formulation for learning the model parameters becomes

$$\max -\frac{1}{2} \ln |C_T| - \frac{1}{2} (\mathbf{y}_T - \boldsymbol{\mu}_T)^T C_T^{-1} (\mathbf{y}_T - \boldsymbol{\mu}_T). \quad (15)$$

As per [32], more robust training can often be achieved using an alternate leave-one-out (LOO) log predictive probability (pseudolikelihood) formulation [30]

$$\begin{aligned} \max \sum_{i=1}^{N_T} \ln p(\mathbf{y}_T^{(i)} | X_T, \mathbf{y}_T^{(-i)}, \mathcal{D}_S, \theta) \\ = \sum_{i=1}^{N_T} \left(-\frac{1}{2} \ln \sigma_{\text{LOO}}^{(i)2} - \frac{(\mathbf{y}_T^{(i)} - \mu_{\text{LOO}}^{(i)})^2}{2\sigma_{\text{LOO}}^{(i)2}} - \frac{1}{2} \ln 2\pi \right) \end{aligned} \quad (16)$$

where the notation $\mathbf{y}_T^{(-i)}$ refers to all target outputs with the exception of the i th data point. The LOO mean $\mu_{\text{LOO}}^{(i)}$ and variance $\sigma_{\text{LOO}}^{(i)2}$ are obtained from the model's predictive distribution at $\mathbf{x}_T^{(i)}$ conditioned on $\{X_T^{(-i)}, \mathbf{y}_T^{(-i)}\}$ and $\mathcal{D}_S$.

We find that for most applications with shallow M (e.g., a single-layer neural network), the full set of Hetero-TGP parameters are satisfactorily learned using off-the-shelf optimizers. When using the log-likelihood formulation for training, its partial derivatives with respect to parameters $\{\theta_k, \sigma_S^2, \sigma_T^2, \lambda\}$ can be computed as shown in [7].

C. Predictive Distribution of the Hetero-TGP Model

The mean ($\tilde{\mu}$) and variance ($\tilde{\sigma}^2$) of the predictive distribution of the Hetero-TGP model at any input $\mathbf{x}_T^{(*)}$ is

$$\tilde{\mu}(\mathbf{x}_T^{(*)}) = \tilde{\mathbf{k}}_*^T (\tilde{\mathbf{K}} + \Lambda)^{-1} \mathbf{y} \quad (17)$$

$$\tilde{\sigma}^2(\mathbf{x}_T^{(*)}) = \tilde{\mathbf{k}}_{**} - \tilde{\mathbf{k}}_*^T (\tilde{\mathbf{K}} + \Lambda)^{-1} \tilde{\mathbf{k}}_* \quad (18)$$

where $\Lambda = \begin{pmatrix} \sigma_S^2 & I_{N_S} & 0 \\ 0 & \sigma_T^2 & I_{N_T} \end{pmatrix}$. It is worth noting that when $\lambda = 0$, which reflects a lack of source-target correlation, we get $\mathbf{K}_{ST} = \mathbf{0}$, and further $\tilde{\mathbf{k}}_* = [\tilde{K}(\mathbf{x}_T^{(*)}, X_S); \tilde{K}(\mathbf{x}_T^{(*)}, X_T)] = [\mathbf{0}; \tilde{K}(\mathbf{x}_T^{(*)}, X_T)]$. In this case, the source-to-target as well as the source-to-source blocks of the matrix $\tilde{\mathbf{K}}$ will have no influence on $\tilde{\mu}(\mathbf{x}_T^{(*)})$ or $\tilde{\sigma}^2(\mathbf{x}_T^{(*)})$, thus reducing the threat of negative transfer as the predictive distribution is affected only by the target data.

IV. GENERALIZED TBO

A pseudocode of the proposed Hetero-TBO is presented in Algorithm 1. The key distinction between Hetero-TBO and conventional BO lies in the incorporation of the Hetero-TGP for probabilistic modeling of the target objective function, thus facilitating the utilization of prior knowledge even in cases where the number of features change across the source and target tasks. Our theoretical analysis shows that this enhancement provides notable advantages with regard to the convergence behavior of the overall algorithm, relative to the standard baseline without transfer learning.

A degree of freedom in the design of Algorithm 1 is the choice of the acquisition function α . The role of α is to serve as a computationally cheap surrogate to the original (expensive) objective function f_T , and is usually configured to offer a balance between exploration and exploitation of the target search space. While a plethora of acquisition functions have been proposed over the years, perhaps the most popularly used is the expected improvement (EI) over the best objective function value ($f_{\min}$) found so far [33]. For a minimization problem, the EI at any previously unevaluated input $\mathbf{x}_T^{(*)}$ is given by

$$\alpha_{EI} = \int_{-\infty}^{\infty} \max \{f_{\min} - f_T^{(*)}, 0\} p(f_T^{(*)} | \mathcal{D}_S, \mathcal{D}_T, \mathbf{x}_T^{(*)}) \times df_T^{(*)}. \quad (19)$$

Another common choice is the upper confidence bound (UCB) measure. Not only is the UCB simple to interpret, but also lends itself well to theoretical analysis [34]. For the case of minimization problems, the UCB at $\mathbf{x}_T^{(*)}$ is

$$\alpha_{UCB} = -\tilde{\mu}(\mathbf{x}_T^{(*)}) + \beta^{1/2} \tilde{\sigma}(\mathbf{x}_T^{(*)}) \quad (20)$$

Algorithm 1 Pseudocode of the Hetero-TBO

Inputs: Computationally expensive objective function f_T ; computational budget; search space $\mathcal{X}_T$; source data set $\mathcal{D}_S$; initial solutions $\mathbf{x}_T^{(1)}, \dots, \mathbf{x}_T^{(N_T)}$

Output: Best solution found

Step 1 - Evaluation of Initial Solutions:

for $i \leftarrow 1$ **to** N_T **do** $y_T^{(i)} \leftarrow f_T(\mathbf{x}_T^{(i)})$

$\mathcal{D}_T \leftarrow \{(\mathbf{x}_T^{(i)}, y_T^{(i)})\}_{i=1}^{N_T}$

while $N_T \leq \text{computational budget}$ **do**

Step 2 - Update the Hetero-TGP model:

$p(f_T | \mathcal{D}_S, \mathcal{D}_T) \leftarrow \text{update model given } \mathcal{D}_T, \mathcal{D}_S$

Step 3 - Optimize α for next solution $\mathbf{x}_T^{(N_T+1)}$:

$\mathbf{x}_T^{(N_T+1)} = \arg\max_{\mathbf{x}_T \in \mathcal{X}_T} \alpha(\mathbf{x}_T; p(f_T | \mathcal{D}_S, \mathcal{D}_T))$

$y_T^{(N_T+1)} \leftarrow f_T(\mathbf{x}_T^{(N_T+1)})$

$\mathcal{D}_T \leftarrow \{\mathcal{D}_T, (\mathbf{x}_T^{(N_T+1)}, y_T^{(N_T+1)})\}$

$N_T \leftarrow N_T + 1$

end

return $\arg\min_{\mathbf{x}_T \in \{\mathbf{x}_T^{(1)}, \dots, \mathbf{x}_T^{(N_T)}\}} y_T$

where β is user-defined. The exploration-exploitation trade-off in the UCB acquisition function originates from its optimism in the face of uncertainty, as captured by the standard deviation term in (20). Reducing the value of β thus tends to reduce the exploration capacity of the algorithm.

A. Regret Bounds of the Hetero-TBO Algorithm

Prior theoretical analyses of BO have defined the *instantaneous regret* of the algorithm at the N th evaluation as $r_N = f(\mathbf{x}^{(N)}) - f(\mathbf{x}^{\text{opt}})$ (for a minimization problem). Then, the resultant *cumulative regret* R_N is defined as the sum of instantaneous regrets over N evaluations, that is $R_N = \sum_{i=1}^N r_i$. Based on these definitions, the cumulative regret bound of a conventional BO algorithm for an objective function f , with a correctly determined GP prior, known additive noise variance (σ_n^2), and a finite search space $\mathcal{X}$, was derived in [35]. In particular, for the case of BO-UCB, it was shown that the regret bound has the form

$$\Pr\{R_N \leq \sqrt{C_1 \beta_N \gamma_N N} \quad \forall N \geq 1\} \geq 1 - \delta. \quad (21)$$

In the above, $C_1 = 8/\log(1 + \sigma_n^{-2})$, $\beta_N = 2 \log(|\mathcal{X}| N^2 \pi^2 / 6\delta)$ where we pick $\delta \in (0, 1)$, and γ_N is the *maximum information gain* after N evaluations, which is defined as

$$\gamma_N = \max_{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)} \in \mathcal{X}} \mathbb{I}([y^{(1)}, \dots, y^{(N)}]; \mathbf{f}). \quad (22)$$

Here, $\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(N)})]$ and $\mathbb{I}(\cdot; \cdot)$ represents the *mutual information* between $\mathbf{f}$ and the observations $y^{(1)}, \dots, y^{(N)}$. Given the basic assumption that $\mathbf{f}$ comes from a multivariate Gaussian distribution (with a covariance matrix $\mathbf{K}$), and that the additive noise is normally distributed, the mutual information is expressed as

$$\mathbb{I}([y^{(1)}, \dots, y^{(N)}]; \mathbf{f}) = \frac{1}{2} \log |I + \sigma_n^{-2} \mathbf{K}|. \quad (23)$$

An immediate corollary of the result in (21) is that as $N \rightarrow \infty$, $R_N/N \rightarrow 0$ with high probability, which implies that the BO-UCB possesses the property of having asymptotically *no-regret*. Moreover, since the cumulative regret is bounded in terms of γ_N , external sources of information that decrease the entropy over target observations will lead to tighter regret bounds, and thus, improve convergence rate.

To this end, the theorem below indicates that the maximum information gain $\tilde{\gamma}_{N_T}(\lambda)$ in the Hetero-TBO algorithm, given source data $\mathcal{D}_S$ with *correctly ascertained* source-target correlation λ , is decreased compared to γ_{N_T} of the no-transfer case. Hence, a tighter regret bound relative to BO-UCB is achieved, which implies that Hetero-TBO should be able to find the optimal solution more quickly.

Theorem 1: The maximum information gain $\tilde{\gamma}_{N_T}(\lambda)$ in the Hetero-TBO algorithm satisfies the relation $\tilde{\gamma}_{N_T}(\lambda) \leq \gamma_{N_T}$, where $|\lambda| \leq 1$ is the source-target correlation under any admissible source feature transformation M . Further, if $|\lambda_a| < |\lambda_b|$, then $\tilde{\gamma}_{N_T}(\lambda_b) \leq \tilde{\gamma}_{N_T}(\lambda_a) \leq \gamma_{N_T}$.

The proof is given in the Appendix. The key message is that as long as there exists some (positive or negative) correlation between the source and target tasks, Hetero-TBO is equipped to adaptively exploit the relationship to lead to faster convergence. The threat of convergence slowdown due to predominantly negative transfer is effectively curtailed. The theorem also tells us that larger the magnitude of source-target correlation (regardless of its sign), the faster the Hetero-TBO algorithm is expected to converge. These deductions provide the necessary scaffolding to support the generality of the Hetero-TBO algorithm. Importantly, the results hold under any (admissible) source feature transformation function M , be it randomly assigned, hand-crafted, or jointly learned.

V. EVALUATING THE HETERO-TGP REGRESSION MODEL

In this section, the predictive accuracy of our proposed transfer learning algorithm is showcased. The practically motivated case study we consider is drawn from the domain of aircraft engine design. Our experimental investigation compares different configurations of the Hetero-TGP model against a standard GP and a *Naive-TGP* approach to cope with changing numbers of features across source and target tasks.

A. Experimental Settings

The GP code for these experiments is obtained from the widely used *gpml* package [36]. In the current implementation of *Hetero-TGP*, we focus on applications characterized by limited labeled data in the target domain, as is always the case in applications of BO to computationally expensive problems. Given limited data, a shallow mapping function M , in the form of a single-layer neural network, is preferred. The rectified linear activation unit is used, introducing nonlinearity in the source feature transformation. The parameters of M are either randomly assigned (leading to a random transformation of the source features), or are jointly learned together with the parameters of the TGP kernel. The case of a hand-crafted mapping function is not considered here as we are primarily interested in the performance achieved when there is no human

TABLE I
LIST OF INPUT FEATURES AND OUTPUT OBJECTIVES FOR THE SOURCE AND TARGET ENGINE SIM DATA SETS

Input Features	Source J85 Turbojet	Target CF6 Turbofan	Output Objectives
Altitude-fit	✓	✓	O1: Net Thrust
Pressure Ratio (Compressor)	✓	×	O2: Fnet/W
Pressure Ratio (Burner)	✓	✓	O3: Fuel Flow
Efficiency (Turbine)	✓	✓	O4: TSFC
A8/A2 (Nozzle)	✓	✓	O5: Core Airflow

in the loop. With regard to the *Naive-TGP* baseline, we shall explain the problem-specific model specifications after the details of the case study are presented.

In all runs, the squared exponential kernel is used. Further, we present results of Hetero-TGP models trained either based on the usual (log) marginal likelihood formulation of (15) or the LOO log predictive probability formulation in (16).

B. Case Study in Aircraft Engine Design

Aircraft engine design is a domain showcasing high complexity due to the presence of mutually interdependent systems that govern overall operation. Consequently, engineers conducting design analyses rely heavily on expensive numerical simulations, due to the lack of availability of closed-form analytical solutions. The data sets we have acquired are generated using the open-source EngineSim (v1.8a) simulator developed by the NASA Glenn Research Center [37]. The simulator is a simple, scalable, and time-economical educational tool designed to model several common aircraft propulsion systems. In an optimization setting, the tool can be used to evaluate the performance of a particular candidate solution (engine configuration) on a set of objectives (e.g., engine efficiency, weight, thrust). The different input features and the five output objectives of interest for the source and target tasks are summarized in Table I. Notice that the source task (J85 turbojet engine) consists of five input features, while in the target task (CF6 turbofan engine) one design parameter is discarded (i.e., it consists of four input features). The change in the number of input features, and the potential impact of this change on the predictive accuracy of transfer learning, illustrates the source-target heterogeneity considered in this article.

Given the description above, the *Naive-TGP* model has been designed keeping in mind the partial knowledge of the discarded feature. In particular, *Naive-TGP* represents a simplified-transfer learning approach in which the missing feature is first filled-in with the mean value of the corresponding feature in the source data [9], as a common form of data imputation. Thereafter, the source and target data are merged, and the standard GP is applied to the combined data set. The adaptive parameter λ does not occur in the *Naive-TGP*.

In the experiments, the training data for the target task contains only seven instances, acquired using Latin hypercube sampling (LHS). On the other hand, 50 samples are collected from the source task. We note that the sizes of the data sets mimic a scenario typically faced by design engineers in the

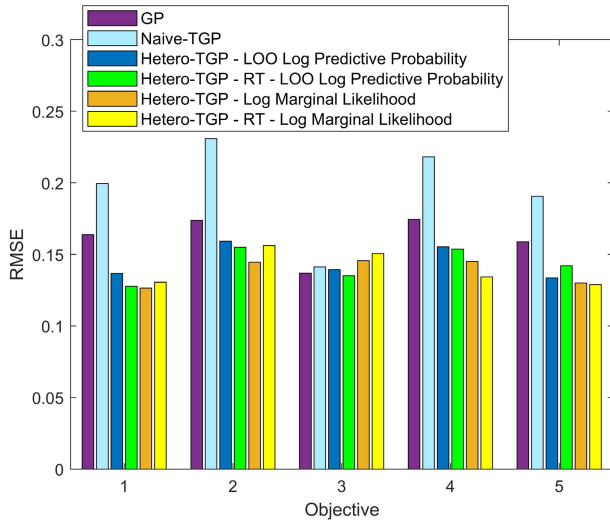


Fig. 1. RMSE comparison between GP, Naive-TGP, and Hetero-TGP for the EngineSim data set (see Table I for list of output objectives). Different configurations of Hetero-TGP are examined, namely the LOO log predictive probability and the log-likelihood formulations, as well as a learned or randomized source feature transformation M . The random transformation case is denoted as RT. All results are averaged over 20 runs.

early project development phase; specifically, building a target regression model with limited data (e.g., during the initial stages of a design optimization process), and augmenting the model with related data from a previously explored source task. Finally, the performance of the trained models are tested on unseen data comprising 1000 samples drawn from the target CF6 turbofan engine simulation.

As depicted in Fig. 1, in 4 out of 5 of the output objectives, Hetero-TGP led to visibly superior performance in terms of the root-mean-square error (RMSE) as compared to standard GP (trained using only the target data) and Naive-TGP. Notably, due to the lack of an adaptation mechanism in Naive-TGP, it suffers from severe negative transfers, which makes its predictive accuracy significantly worse than that of the standard GP. These results showcase the importance of the parameter λ in enabling adaptive-transfer learning. For output O3 however, the performance of all models is comparable, suggesting a lack of useful information gained from the source.

Aside from the above, it was observed that the performance of the Hetero-TGP model is sensitive to parameter initializations when using the log-likelihood formulation for training. Therefore, in the black-box optimization experiments that follow, the LOO log predictive probability formulation is used throughout for model training.

VI. EVALUATING THE HETERO-TBO ALGORITHM

In this section, we showcase the efficacy of the Hetero-TBO algorithm compared to conventional BO on a series of synthetic functions, namely, the multimodal *Griewank*, *Rastrigin*, *Six-Hump Camel*, and the unimodal *Sphere*. Mathematical expressions of these functions can be found in [38].

A. Experimental Settings

The experiments are carried out using both EI and UCB acquisition functions. For the UCB, β is set to 0.1. With

TABLE II
PARAMETER SETTINGS OF THE GA USED FOR MAXIMIZING THE ACQUISITION FUNCTION IN HETERO-TBO AND BO

Parameter	Setting
Number of GA iterations	500
Size of GA population	20
Mutation probability	$1/d_T$
SBX and polynomial mutation index	10
Crossover probability	0.9

TABLE III
HETEROGENEOUS SOURCE AND TARGET SEARCH SPACE CONFIGURATIONS IN THE PEDAGOGICAL EXAMPLES. RANDOM DIMENSIONS REPRESENT THE NUMBER OF EXTRA FEATURES THAT HAVE BEEN INSERTED IN THE SOURCE BUT HAVE NO EFFECT ON ITS OBJECTIVE FUNCTION

Function Labels	Source (S)		Target (T)
	Original Dimensions	Random Dimensions	Dimensionality (d_T)
P1: <i>Sphere</i> ($d_S < d_T$)	5	0	7
P2: <i>Griewank</i> ($d_S < d_T$)	5	0	7
P3: <i>Six-Hump Camel</i> ($d_S > d_T$)	2	2	2
P4: <i>Rastrigin</i> ($d_S > d_T$)	5	2	5
P5: <i>Rastrigin</i> (S), <i>Sphere</i> (T)	8	0	10
P6: <i>S-H Camel</i> (S), <i>Griewank</i> (T)	2	0	5

regard to the GP kernel and the mapping function M , settings as in Section V are employed. In all experiments, the target optimization starts with a sample of five inputs generated via LHS. The optimization of the acquisition function in both Hetero-TBO and conventional BO is conducted using a simple genetic algorithm (GA) with the following evolutionary operators: simulated binary crossover (SBX) [39], polynomial mutation [40], and elitist selection. Further details of the GA are in Table II. We note that the standard GP-based BO is a natural precedent for comparison as it continues to be the prototypical probabilistic model in BO [41]. Moreover, since our proposed transfer learning algorithm is designed based on the fundamentals of GPs, the comparison is fair given all other experimental settings are kept consistent.

Not limiting to BO, we also compare against *random search*, which has shown its efficacy over grid search for black-box hyperparameter tuning of machine learning models [42]. To account for randomization, all results presented in this section are averaged over 30 independent runs, with the assumption that each run has a limited computational budget.

B. Experimental Settings and Results on Synthetic Functions

The basic problem specifications are summarized in Table III. The problem settings are organized to represent three different experimental categories. In the first four experiments, the source and the target objective functions are essentially the same (such that there exists a definite source-target correlation for Hetero-TBO to exploit). The difference between tasks originates from their differing search space dimensionality. In the first two experiments (i.e., in **P1** and **P2**), the target task has two extra features; that is, $d_S < d_T$. In addition, all target features have been

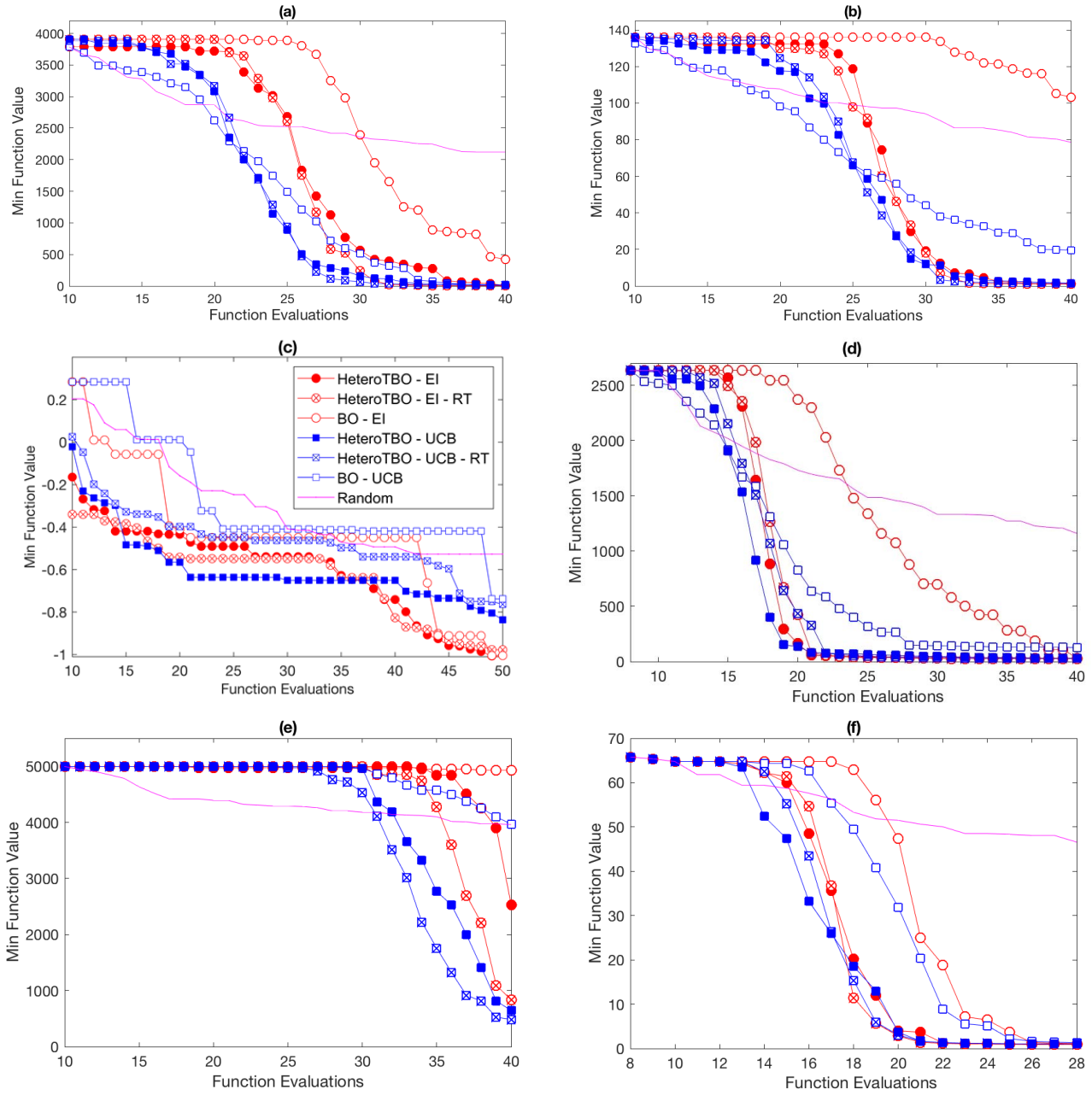


Fig. 2. Comparison of convergence trends between Hetero-TBO, conventional BO, and random search on synthetic benchmarks. Results are presented for both UCB and EI acquisition functions, and are averaged over 30 independent runs. Results where Hetero-TBO uses a randomized source feature transformation are denoted with RT. (a) **P1**, (b) **P2**, (c) **P3**, (d) **P4**, (e) **P5**, and (f) **P6**. Specifications in Table III.

translated relative to the source (by ten units along each dimension for *Sphere*, and 50 units along each dimension for *Griewank*). Feature translation not only breaks the direct correspondence between the subset of overlapping features of the source and target tasks, but also shifts the distribution of sampled solutions. In contrast, in **P3** and **P4**, we insert two extra *random* features into the source task, so that $d_S > d_T$. It is noted that the random features have no effect on the objective value, and simply attempt to confound the Hetero-TBO algorithm. Finally, in the last two examples (i.e., in **P5** and **P6**) we not only have $d_S \neq d_T$, but also the source and target objective functions are different (i.e., $f_S \neq f_T$).

As such, in **P5** and **P6**, there is little direct correspondence between the source and target features.

In Fig. 2, we see that for all problems, Hetero-TBO variants consistently lead to faster convergence trends as compared to the baseline algorithms. Overall, random search turns out to be ineffective across all the benchmarks. Although it makes better progress at the very start, it is eventually outperformed by Hetero-TBO and BO as these algorithms accumulate more data to overcome their cold start. It is worth noting that Hetero-TBO, by virtue of its transfer learning capability, is in general faster to overcome the cold start problem as compared to conventional BO. The effect of cold start is particularly

pronounced for **P5** [see Fig. 2(e)] as the dimensionality of the problem is relatively large ($d_T = 10$) compared to the rest, implying that much more data is needed before informative regression models can be built.

To put the speed-up achieved by Hetero-TBO into perspective, observe that the minimum function value reached by BO-EI at 40 evaluations is achieved by Hetero-TBO-EI in much less than 30 evaluations for **P2** and **P4**. This equates to a saving of more than 25%, indicating the potential to reduce several hours (or even days) of computation time as each evaluation becomes increasingly time consuming (e.g., in engineering design applications where a single evaluation based on a finite element simulation could take several hours [9]). Through this set of six experiments (**P1–P6**), we provide evidence that Hetero-TBO can indeed be effective when $d_S < d_T$ [see Fig. 2(a) and (b)], when $d_S > d_T$ [see Fig. 2(c) and (d)], as well as when $f_S \neq f_T$ [see Fig. 2(e) and (f)]; thus demonstrating its generality. The experimental results support our theoretical claim that the Hetero-TBO algorithm can adaptively transfer knowledge across black-box optimization tasks, while reducing the impact of harmful negative transfer.

Although Hetero-TBO with random assignment of the parameters of the mapping function M is found to work well for the synthetic benchmarks, a learned source feature transformation is expected to provide more reliable performance on average. For this reason, in all subsequent experiments on expensive real-world problems, we consider M to be jointly learned with the TGP kernel parameters. We call on the typical assumption in BO that the cost of building the regression model is negligible when compared to the cost of a single evaluation of the computationally expensive objective.

VII. PRACTICAL OPTIMIZATION EXEMPLARS

A. Accelerating AutoML via Knowledge Transfer

Determining the optimal set of hyperparameters for a machine learning model can be a time consuming affair, as it typically entails a rigorous cross-validation procedure for approximating the out-of-sample predictive performance of a candidate model. Yet, an appropriate selection of these hyperparameters can often be the key to achieving state-of-the-art performance. To this end, one of the fundamental goals of AutoML is to fully automate the process of hyperparameter tuning, without the need for a human in the loop.

In this experiment, we demonstrate the efficacy of Hetero-TBO with regard to harnessing latent correlations between the hyperparameter configurations of different neural network architectures. In particular, we consider the target task to be the hyperparameter tuning exercise of a convolutional neural network (CNN), comprising one convolutional layer with max pooling followed by two densely connected layers, applied to the MNIST handwritten digits data set. 50 source data samples are taken from hyperparameter configurations of a CNN applied to the related USPS digits task. The network utilizes dropout for regularization, the rectifier linear unit activation function, and weight updates calculated via stochastic gradient descent with momentum.

TABLE IV
UPPER AND LOWER BOUNDS OF HYPERPARAMETERS IN
THE SOURCE AND TARGET NEURAL NETWORKS

Features	Source	Target	Lower Bound	Upper Bound
Learning rate	✓	✓	10^{-5}	0.5
Momentum	✓	✓	0	1
L2 regularization	✓	✓	0	0.1
Batch size	✓	✓	32	256
Dropout probability	✓	✓	0	0.9
Nodes in fully-connected layers	✓	×	10	200

The network hyperparameters of interest are listed in Table IV, together with the bounds of their respective search range. In the target optimization task, one of the hyperparameters, namely the number of hidden nodes in the fully connected layers, is fixed to 100 (as a means of restraining model complexity). Thus, a heterogeneous transfer setting occurs with $d_S > d_T$. Given a candidate set of hyperparameters, the target CNN is trained for a maximum of three epochs.

While both MNIST and USPS cater to handwritten digit recognition, there are some obvious difference between them, for example, image resolution, slanted numbers, padding, and thicker lines. Nevertheless, we observe from Fig. 3(a) that the Hetero-TBO approach is capable of exploiting the underlying similarities between the two learning problems to accelerate the search for good hyperparameter configurations as compared to the baseline algorithms. To elaborate, for achieving a minimum prediction error of 0.014, BO-EI consumes an average of 30 function evaluations, while Hetero-TBO-EI takes only 20 evaluations. This equates to approximately a 33% saving.

B. Case Study in Advanced Material Manufacturing

In the domain of engineering design, products and processes are routinely subjected to gradual modifications over time. As such, iterative design exercises are likely to share underlying commonalities, giving rise to the scope for knowledge transfer. It is further noted that high-fidelity physics simulations, which are essential elements of engineering analyses, are typically computationally expensive, consuming several minutes, hours, or even days for a single evaluation; thus, forming a major bottleneck in the design cycle. For this reason, there is indeed much to be gained by reducing the number of function evaluations needed to achieve optimal or near-optimal designs.

To this end, we demonstrate the efficacy of Hetero-TBO on a real-world *simulation-based* engineering design optimization problem. In particular, we consider two distinct members of a family of rigid tool liquid composite molding (LCM) processes, namely, resin transfer molding (RTM), and injection/compression liquid composite molding (I/C-LCM). These processes are popular methodologies for the manufacture of high strength-to-weight ratio composite parts in the aerospace and automotive industries.

In this case study, the manufacture of a particular glass-fiber and epoxy composite part is considered via the RTM and

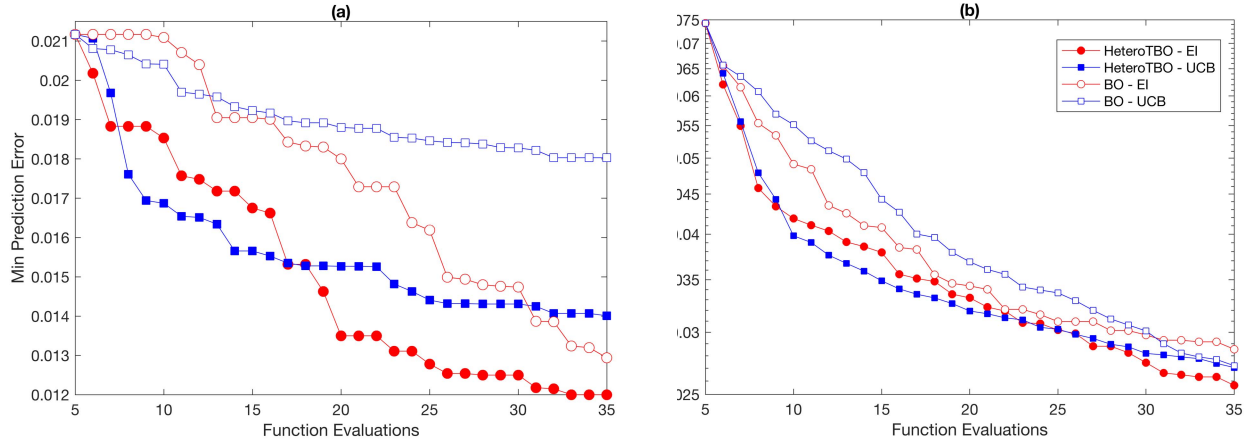


Fig. 3. Convergence trends of Hetero-TBO and BO for (a) automatic hyperparameter tuning of a machine learning model and (b) advanced material manufacturing. Results are averaged over ten runs in the computationally expensive case.

TABLE V
UPPER AND LOWER BOUNDS OF DESIGN VARIABLES OF THE
I/C-LCM (SOURCE) AND RTM (TARGET) PROCESSES

Features	Source	Target	Lower Bound	Upper Bound
$V_{closure}$	✓	✓	1 mm/min	10mm/min
P_{inj}	✓	✓	1 MPa	10 MPa
T_{mold}	✓	✓	293 K	348 K
T_{resin}	✓	✓	293 K	348 K
$V_{closure}^{final}$	✓	×	1 mm/min	10mm/min
H_{inj}	✓	×	0.8 cm	1 cm

the I/C-LCM processes. The target task is the *single phase* RTM process that involves four design variables, namely: 1) speed of mold closure $V_{closure}$; 2) resin injection pressure P_{inj} ; 3) preheated mold temperature T_{mold} ; and 4) preheated resin temperature T_{resin} . For Hetero-TBO, source data is drawn from a previous design exercise for the relate *two-phase* I/C-LCM process. Unlike RTM, the overall I/C-LCM cycle has six design variables, four of which are overlapping with RTM. The new features added in the case of I/C-LCM are: 1) mold cavity thickness during resin injection H_{inj} and 2) velocity of final mold closure $V_{closure}^{final}$. The design variables for the two tasks, and their respective search ranges, are summarized in Table V; for further details the reader may refer to [10].

In this article, the objective function is a weighted aggregation of the *process time* (which determines the throughput) and the *peak internal force* (which determines the set up and running cost of peripheral equipment). We note that the inclusion of two additional input features in I/C-LCM generally leads to resin flow behaviors that are seemingly different from that of RTM. However, there continue to exist known correlations between the induced flow behaviors and the input features that are preserved across both manufacturing processes. For instance, increasing the resin injection pressure always tends to decrease process time but at the cost of increasing the total internal force. Similarly, an increase in mold or resin temperature has the effect of easing resin flow (thus reducing process time) due to drop in resin viscosity, regardless of whether the manufacturing process is RTM or I/C-LCM. Although

the extra features in I/C-LCM do introduce some outward variations in the flow behavior, these underlying correlations remain. It is the presence of such commonalities that are exploited by the Hetero-TBO algorithm when transfer learning from I/C-LCM to RTM, in turn accelerating the process design optimization.

The obtained experimental result is reported in Fig. 3(b), where we find that the knowledge transfer of Hetero-TBO provides an impetus to the convergence rates. For example, to achieve an aggregated and normalized objective function value of 0.04 Hetero-TBO-UCB takes an average of ten function evaluations, whereas BO-UCB (which starts the search from scratch) consumes 17 function evaluations. This indicates a saving of approximately 41%.

VIII. CONCLUSION

In this article, we put forward a generalization of TBO to problems characterized by source-target heterogeneity. To this end, we proposed a novel Hetero-TGP model that can cope with scenarios where (in addition to $f_S \neq f_T$) the number of features could change across the source and target tasks. The Hetero-TGP model is based on joint learning of a new feature representation for the source data, one that is better aligned with the features of the target task of interest. A tightening of the cumulative regret bounds of the resultant Hetero-TBO algorithm was revealed, guaranteeing faster convergence as opposed to conventional BO. Moreover, our theoretical result indicated that even a random source feature transformation can be afforded in Hetero-TBO, reducing the complexity of model learning without overturning the positive benefits of knowledge transfer. This outcome is a consequence of the modified covariance function of Hetero-TGP, enabling the model to adapt to the degree of correlation between source and target tasks such that the harmful effects of negative transfer are suppressed.

In the experimental studies, the predictive performance of the Hetero-TGP model was first shown to be superior to the standard GP. The shortcoming of a naive implementation of GP-based transfer learning was also shown on a data set representative of aircraft engine design applications. Thereafter, the results for black-box optimization were presented,

demonstrating the efficacy of the Hetero-TBO algorithm on both synthetic test functions and real-world problems. The benefits of incorporating the Hetero-TGP model were highlighted by cost savings of more than 25% (measured in terms of function evaluations) when compared to conventional BO.

In addition to unveiling the utility of knowledge transfer even under source-target heterogeneity, our work sheds light on a number of future research directions. For instance, the Hetero-TGP model presented in this article is restricted to a scalar parameter (λ) for capturing source-target correlation. As was concluded in [29], such an approach may not suffice in capturing more complex relationships between tasks as often arises in transfer learning. Thus, exploring alternatives such as multiple kernel transfer learning, or mixtures of TGP experts, in the context of TBO, is an important future extension of our work. In addition, of significant interest to us is the case where we have a large and growing knowledge based of multiple source data sets, making it challenging to efficiently select the most relevant source for transfer [43], [44]. Successfully resolving this obstacle is expected to advance the widespread adoption of TBO as a fast and scalable approach for computationally expensive black-box optimization.

APPENDIX

A. Proof of Theorem 1

The proof methodology follows [45]. Following (22), the maximum information gain for Hetero-TBO, conditioned on the source data set $\mathcal{D}_S$ and an admissible source input transformation function M , may be written as

$$\tilde{\gamma}_{N_T}(\lambda) = \max_{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N_T)} \in \mathcal{X}_T} \mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T | \mathcal{D}_S, M). \quad (24)$$

Given the conditional target covariance matrix C_T in (14), the conditional mutual information between $\mathbf{f}_T$ and $[y_T^{(1)}, \dots, y_T^{(N_T)}]$ takes the form

$$\begin{aligned} \mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T | \mathcal{D}_S, M) \\ = \frac{1}{2} \log |I + \sigma_T^{-2} (K_{TT} - K_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K_{ST})|. \end{aligned} \quad (25)$$

Based on the result in [7], $\tilde{K} = \begin{pmatrix} K_{SS} & K_{ST} \\ K_{TS} & K_{TT} \end{pmatrix}$ is PSD for $|\lambda| \leq 1$. This sufficiency condition holds for any valid kernel $k(\cdot, \cdot)$ and admissible mapping function M . Therefore, from the Schur complement theorem, it follows that the matrix $K_{TT} - K_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K_{ST}$ is also PSD. Further, since $K_{ST} = K_{TS}^T$, $K_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K_{ST}$ is indeed PSD.

Given the above, we invoke the Minkowski determinant inequality which states that for PSD matrices A and B , $|A + B| \geq |A| + |B| \geq |A|$. In the no-transfer case, mutual information $\mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T) = (1/2) \log |I + \sigma_T^{-2} K_{TT}|$. By setting $A = I + \sigma_T^{-2} (K_{TT} - K_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K_{ST})$ and $B = \sigma_T^{-2} K_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K_{ST}$ in the Minkowski inequality, we get

$$\begin{aligned} \mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T | \mathcal{D}_S, M) \\ \leq \mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T). \end{aligned} \quad (26)$$

Thus, for any set $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N_T)} \in \mathcal{X}_T$, $\tilde{\gamma}_{N_T}(\lambda) \leq \gamma_{N_T}$.

Next, consider $K_{ST} = \lambda K'_{ST}$, such that K'_{ST} is independent of λ . We may then write

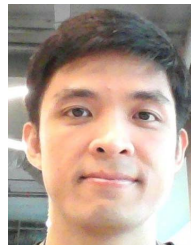
$$\begin{aligned} \mathbb{I}([y_T^{(1)}, \dots, y_T^{(N_T)}]; \mathbf{f}_T | \mathcal{D}_S, \lambda, M) \\ = \frac{1}{2} \log |I + \sigma_T^{-2} (K_{TT} - \lambda^2 K'_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K'_{ST})|. \end{aligned} \quad (27)$$

Since, $A(\lambda) = I + \sigma_T^{-2} (K_{TT} - \lambda^2 K'_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K'_{ST})$, we have $A(\lambda_a) = A(\lambda_b) + \sigma_T^{-2} (\lambda_b^2 - \lambda_a^2) K'_{TS} (K_{SS} + \sigma_S^2 I)^{-1} K'_{ST}$. Assuming $|\lambda_a| < |\lambda_b|$, it follows by once again applying the Minkowski determinant inequality that $|A(\lambda_b)| \leq |A(\lambda_a)|$, which implies the latter statement of the theorem, that is, $\tilde{\gamma}_{N_T}(\lambda_b) \leq \tilde{\gamma}_{N_T}(\lambda_a)$.

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Bayesian optimization, and evolutionary optimization, to facilitate rapid improvements in application areas, such as engineering design and robotics.

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